user-level tools for OpenVZ (*yum install vzctl vzquota*); *vzctl* is used to perform different tasks on VPSs such as *Start, Stop, Create, Destroy, Set parameters*, etc, while *vzquota* is used to manage the VPS quotas.
7. Edit */etc/sysctl.conf* and add or modify the following settings:
8. Load the new settings with *sysctl –p*
9. Open */etc/vz/vz.conf* and set *NEIGHBOUR_DEVS* to *all*.
10. SELinux needs to be disabled; edit */etc/sysconfig/selinux* and set the value of *SELINUX* to *disabled*.
11. Reboot the system. You can check the release with *uname –r* and should get something like *2.x.xx-xxxstabxxx.x*.
12. OpenVZ is installed in a way that it is possible to boot the system either with OpenVZ support or without it. At present, it is not possible to create VPSs. Different VPSs can run different versions of Linux. A VPS is based on a specific OS template; OS templates are packages available with OpenVZ. You need to install the corresponding OS template in OpenVZ to create a VPS. After you install at least one OS template, you can create any number of VPSs with the help of standard OpenVZ utilities, and can configure their network, and work with these VPSs as you work with fully functional Linux servers. First download a CentOS template:

```
wget http://download.openvz.org/template/precreated/centos-
6-x86_64.tar.gz
```

13. Copy the downloaded template into */vz/template/cache*.
14. Each VPS must have its own unique ID (here, we used 121) and create the VPS with:

```
vzctl create 121 --ostemplate centos-6-x86_64 --config basic
```

15. Set a hostname and IP address for the VPS:

```
vzctl set 121 --hostname testvps.com --save
vzctl set 121 --ipadd 192.168.0.201 –save
```

16. You can now start the VPS with *vzctl start 121*; stop it with *vzctl stop 121* and restart it with *vzctl restart 121*.

## EasyVZ: OpenVZ management GUI

EasyVZ is a GUI management console for OpenVZ. It lets you easily create, destroy, manage and monitor VPSs. The pre-requisites for using it are:
1. You have to have an OpenVZ-enabled kernel running.
2. All OpenVZ utilities need to be installed.
3. To create new VPSs, you need templates installed in */vz/template/cache*.

The source distribution contains two directories; the *backend* directory contains the server source code. Start the server on the OpenVZ node that you intend to manage:

```
cd backend/
python server.py
```

The client can be run on the same machine:

```
cd gui/
python ezvz.py
```

## Monitoring system resource consumption

It is possible to check the system resource statistics from within a VPS, which allows you to understand what particular resource limits are preventing an application from starting. These statistics report the current and maximum resources consumption for the running VPS, and can be obtained from the */proc/user_beancounters* file:

```
vzctl exec 101 cat /proc/user_beancounters
```

## Monitoring memory consumption

Users can monitor memory parameters for the hardware node and for particular VPSs:

```
vzmemcheck –v
```

The –v option is used to display the memory information for each VPS.

## Enabling VPN for a VPS

A Virtual Private Network allows you to establish a secure network connection over an insecure public network. It is possible to set up a VPN for a separate VPS via the TUN/TAP device.
1. The *tun* module needs to be loaded before OpenVZ is started; you can load it with:

```
lsmod | grep tun or modprobe tun
```

2. Allow the VPS to use the TUN/TAP device:

```
vzctl set 121 --devices c:10:200:rw –save
```

3. Create the subsequent device inside the VPS and set the proper permissions. END

### References


[0] http://en.wikipedia.org/wiki/Comparison_of_platform_virtual_machines
[1] http://en.wikipedia.org/wiki/Network_virtualization
[2] http://en.wikipedia.org/wiki/Storage_Virtualization
[3] http://en.wikipedia.org/wiki/Memory_virtualization
[4] http://wiki.openvz.org/Main_Page
[5] http://wiki.openvz.org/VPS_vs_Dedicated
[6] http://wiki.openvz.org/Use_cases
[7] http://sourceforge.net/projects/easyvz/
[8] http://download.openvz.org/doc/OpenVZ-Users-Guide.pdf



**By: Mitesh Soni**

The author is a Technical Lead at iGATE. He is in the Cloud Services (Research & Innovation) Group and loves to write about new technologies. Blog: http://clean-clouds.com